

Algebraic Techniques

Week 2C

Overview

Indices

Key Ideas - Index Notation

- Positive Index Rules

Key Idea – Index Notation

Basic indices involve **repeated multiplication**. However, the topic becomes more complicated when division and taking roots are represented by indices. In this chapter we will look at how indices are written and state the rules which are used to simplify them.

Discussion Question:

What does 2^6 mean?

$$2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

Indices **represent repeated multiplication**, if a is a real number and n is a natural number then the **general definition for index notation** is:

$$a^n = \overbrace{a \times a \times \cdots \times a}^{n \text{ terms}}.$$

- a is called the **base**
- n is called the **index** of a (or *power to which a is raised, or exponent on a*)

Note: $a^1 = a$

Examples:

The expression 6×6 :

- Can be written in index notation as 6^2 , the index is 2 and the base is 6.
- 6^2 could be read as “6 to the second power”, “6 raised to the power of 2” or simply “6 squared”.

The expression $a \times a \times a$:

- Can be written in index notation as a^3 , the index is 3 and the base is a .
- In words, a^3 could be read as “ a to the third power”, “ a raised to the power of 3” or simply “ a cubed”.

Examples:

$$10^2 = 10 \times 10 = 100$$

$$5^3 = 5 \times 5 \times 5 = 125$$

$$2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$$

$$(2x)^3 = 2x \times 2x \times 2x = 2 \times 2 \times 2 \times x \times x \times x = 8x^3$$

$$x^4 \times y \times x \times y = x \times x \times x \times x \times y \times x \times y = x \times x \times x \times x \times x \times y \times y = x^5 y^2$$

$$\left(-\frac{1}{3}\right)^3 = -\frac{1}{3} \times -\frac{1}{3} \times -\frac{1}{3} = -\frac{1}{27}$$

Exercises:

Expand or simplify the indices below:

$$(a) 3^3 = 3 \times 3 \times 3 = 27$$

$$(b) (2a)^4 = 2a \times 2a \times 2a \times 2a = 16a^4$$

$$(c) 2a^4 = 2 \times a \times a \times a \times a$$

$$(d) \left(-\frac{1}{2}\right)^4 = \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) = \frac{1}{16}$$

$$(e) \left(\frac{2x}{y}\right)^3 = \frac{2x}{y} \times \frac{2x}{y} \times \frac{2x}{y} = \frac{8x^3}{y^3}$$

Key Idea – Positive Index Rules

Before introducing the rules for Indices, it is important to understand where they have come from.

Discussion Questions:

How can we simplify the two expressions $2^2 \times 2^3$ and $a^2 \times a^3$?

$$2^2 \times 2^3 = 2 \times 2 \times 2 \times 2 \times 2 = 2^5$$

$$a^2 \times a^3 = a \times a \times a \times a \times a = a^5$$

Can you write a general rule for multiplication of indices?

$$a^m \times a^n = a^{m+n}$$

Would this rule apply for $10^3 \times 2^5$?

No the base needs to be the same

Rule 1: Multiplying powers of the same number or variable

To understand the rule, we can expand each power into repeated multiplication and observe the pattern.

$$\begin{aligned}a^2 \times a^3 &= (a \times a) \times (a \times a \times a) \\&= a \times a \times a \times a \times a \\&= a^5 \\&= a^{2+3}\end{aligned}$$

More generally, for any positive integers m and n ,

$$a^n \times a^m = \overbrace{a \times a \cdots \times a}^{n \text{ terms}} \times \overbrace{a \times a \cdots \times a}^{m \text{ terms}} = \overbrace{a \times a \cdots \times a}^{n+m \text{ terms}} = a^{n+m}.$$

$$\text{Rule 1: } a^n \times a^m = a^{n+m}$$

Examples:

$$\begin{aligned}3^3 \times 3^4 &= 3^{3+4} \\&= 3^7\end{aligned}$$

$$\begin{aligned}7^5 \times 7^{23} &= 7^{5+23} \\&= 7^{28}\end{aligned}$$

$$\begin{aligned}8 \times 16 &= 2^3 \times 2^4 \\&= 2^{3+4} \\&= 2^7\end{aligned}$$

$$\begin{aligned}x^3 \times x^2 &= x^{3+2} \\&= x^5\end{aligned}$$

Exercises:

Simplify the following, writing each answer in index notation:

$$(a) \ 2^3 \times 2^4 = 2^{3+4} \\ = 2^7$$

$$(b) \ y \times 3y^3 = y^1 \times 3 \times y^3 \\ = 3y^{1+3} \\ = 3y^4$$

Rule 2: Dividing powers of the same number or variable.

Discussion Questions:

How can we simplify the two expressions $2^5 \div 2^2$ and $a^5 \div a^2$?

$$2^5 \div 2^2 = \frac{2^5}{2^2} = \frac{\cancel{2} \times \cancel{2} \times 2 \times 2 \times 2}{2 \times \cancel{2}} = 2^3$$

$$a^5 \div a^2 = \frac{a^5}{a^2} = \frac{\cancel{a} \times \cancel{a} \times a \times a \times a}{\cancel{a} \times \cancel{a}} = a^3$$

Can you write a general rule for division of indices?

$$a^m \div a^n = a^{m-n}$$

Rule 2: Dividing powers of the same number or variable.

To understand the rule, we can expand each power into repeated multiplication and observe the pattern.

$$\begin{aligned}a^5 \div a^2 &= \frac{a^5}{a^2} = \frac{\cancel{a} \times \cancel{a} \times a \times a \times a}{\cancel{a} \times \cancel{a}} \\&= a^3 \\&= a^{5-2}\end{aligned}$$

More generally, for any positive integers m and n ,

$$a^n \div a^m = \frac{\overbrace{a \times a \cdots \times a}^{n \text{ terms}}}{\underbrace{a \times a \cdots \times a}_m} = \overbrace{a \times a \cdots \times a}^{n-m \text{ terms}} = a^{n-m}.$$

$$\textbf{Rule 2: } a^n \div a^m = a^{n-m}$$

Examples:

$$\begin{aligned}2^5 \div 2^3 &= 2^{5-3} \\&= 2^2\end{aligned}$$

$$\begin{aligned}a^5 \div a^2 &= a^{5-2} \\&= a^3\end{aligned}$$

$$\begin{aligned}\frac{6q^8}{24q^6} &= \frac{1}{4}q^{8-6} \\&= \frac{q^2}{4}\end{aligned}$$

Exercises:

Simplify the following, writing each answer in index notation:

$$(a) \frac{3^4}{3^2} = 3^{4-2} = 3^2$$

$$(b) \frac{4b^8}{12b^5} = \frac{4}{12} b^{8-5} = \frac{1}{3} b^3 = \frac{b^3}{3}$$

$$(c) (x^2y^7) \div (xy^4) = \frac{x^2y^7}{xy^4} \\ = x^{2-1}y^{7-4} = xy^3$$

Rule 3: Powers of numbers or variables multiplied

Discussion Questions:

How can we simplify $(ab)^3$?

$$\begin{aligned}(ab)^3 &= (ab) \times (ab) \times (ab) \\ &= a \times b \times a \times b \times a \times b \\ &= a^3 b^3\end{aligned}$$

Can you write a general rule for this expression?

$$(ab)^m = a^m b^m$$

Rule 3: Powers of numbers or variables multiplied

To understand the rule, we can expand the power into repeated multiplication and observe the pattern.

$$\begin{aligned}(a \times b)^3 &= (a \times b) \times (a \times b) \times (a \times b) \\ &= (a \times a \times a)(b \times b \times b) \\ &= a^3 \times b^3\end{aligned}$$

More generally, for any positive integer n ,

$$\begin{aligned}(a \times b)^n &= \overbrace{(a \times b) \times (a \times b) \times \cdots \times (a \times b)}^{n \text{ pairs of brackets}} \\ &= \overbrace{a \times a \times \cdots \times a \times b \times b \times \cdots \times b}^{2n \text{ terms}} \\ &= \overbrace{a \times a \times \cdots \times a}^{n \text{ terms}} \times \overbrace{b \times b \times \cdots \times b}^{n \text{ terms}} \\ &= a^n \times b^n\end{aligned}$$

$$\text{Rule 3: } (ab)^n = a^n \times b^n$$

Note: In algebra we write $a^n \times b^n$ as $a^n b^n$ and we write $a \times b$ as ab

Examples:

$$\begin{aligned}(2x)^5 &= (2 \times x)^5 \\ &= 2^5 x^5 \\ &= 32x^5\end{aligned}$$

$$\begin{aligned}(3y)^4 &= 3^4 \times y^4 \\ &= 81y^4\end{aligned}$$

$$\begin{aligned}(abc)^2 &= (a \times b \times c)^2 \\ &= a^2 b^2 c^2\end{aligned}$$

$$\begin{aligned}35^p &= (5 \times 7)^p \\ &= 5^p \times 7^p\end{aligned}$$

Rule 4: Powers of numbers or variables divided

Discussion Questions:

How can we simplify $\left(\frac{a}{b}\right)^3$?

$$\begin{aligned}\left(\frac{a}{b}\right)^3 &= \left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \\ &= \frac{a \times a \times a}{b \times b \times b} \\ &= \frac{a^3}{b^3}\end{aligned}$$

Can you write a general rule for this expression?

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

Rule 4: Powers of numbers or variables divided

To understand the rule, we can expand the power into repeated multiplication and observe the pattern.

$$\begin{aligned}\left(\frac{a}{b}\right)^3 &= \left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \\ &= \frac{a \times a \times a}{b \times b \times b} \\ &= \frac{a^3}{b^3}\end{aligned}$$

More generally, for any positive integer n ,

$$\begin{aligned}\left(\frac{a}{b}\right)^n &= \overbrace{\left(\frac{a}{b}\right) \times \left(\frac{a}{b}\right) \times \cdots \times \left(\frac{a}{b}\right)}^{n \text{ pairs of brackets}} \\ &= \frac{\overbrace{a \times a \times \cdots a}^{n \text{ terms}}}{\overbrace{b \times b \times \cdots b}^{n \text{ terms}}} \\ &= \frac{a^n}{b^n}\end{aligned}$$

$$\textbf{Rule 4: } \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \text{ (provided } b \neq 0\text{)}$$

Examples:

$$\left(\frac{x}{y}\right)^5 = \frac{x^5}{y^5}$$

$$\begin{aligned}\left(-\frac{1}{2}\right)^3 &= \frac{(-1)^3}{2^3} \\ &= \frac{-1}{8}\end{aligned}$$

$$\begin{aligned}\left(\frac{2x}{y}\right)^3 &= \frac{(2x)^3}{y^3} \\ &= \frac{2^3 \times x^3}{y^3} && \text{(combining rule 3 and 4)} \\ &= \frac{8x^3}{y^3}\end{aligned}$$

Rule 5: Powers of powers

Discussion Questions:

How can we simplify $(a^2)^3$?

$$\begin{aligned}(a^2)^3 &= a^2 \times a^2 \times a^2 \\ &= a^{2+2+2} \\ &= a^6\end{aligned}$$

Can you write a general rule for this expression?

$$(a^m)^n = a^{m \times n}$$

Rule 5: Powers of powers

To understand the rule, we can expand the power into repeated multiplication twice and observe the pattern.

$$\begin{aligned}(a^2)^3 &= (a \times a)^3 \\ &= (a \times a) \times (a \times a) \times (a \times a) \\ &= a \times a \times a \times a \times a \times a \\ &= a^6 \\ &= a^{2 \times 3}.\end{aligned}$$

More generally, for any positive integers m and n ,

$$\begin{aligned}(a^n)^m &= \overbrace{(a \times a \times \dots \times a)}^{n \text{ terms}}^m \\ &= \overbrace{\overbrace{(a \times a \times \dots \times a)}^{n \text{ terms}} \times \overbrace{(a \times a \times \dots \times a)}^{n \text{ terms}} \times \dots \times \overbrace{(a \times a \times \dots \times a)}^{n \text{ terms}}}^{m \text{ lots of } n \text{ terms}} \\ &= \overbrace{(a \times a \times \dots \times a)}^{nm \text{ terms}} \\ &= a^{nm}\end{aligned}$$

$$\text{Rule 5: } (a^n)^m = a^{n \times m} = a^{nm}$$

Examples:

$$\begin{aligned}(2^2)^3 &= 2^{2 \times 3} \\ &= 2^6 \\ &= 64\end{aligned}$$

$$\begin{aligned}(b^2)^p &= b^{2 \times p} \\ &= b^{2p}\end{aligned}$$

$$\begin{aligned}(-2x^2)^3 &= (-2)^3 \times (x^2)^3 \\ &= -8 \times x^{2 \times 3} && \text{(combining rule 3 and 5)} \\ &= -8x^6\end{aligned}$$

Exercises:

Simplify the following, writing each answer in index notation:

$$\begin{aligned}\text{(a)} \quad (2x)^4 \times x^3 &= 2^4 x^4 \times x^3 \\ &= 16 x^{4+3} \\ &= 16 x^7\end{aligned}$$

$$\begin{aligned}\text{(b)} \quad (-3a^3b^4)^3 &= (-3)^3 \times (a^3)^3 \times (b^4)^3 \\ &= -27 a^{3 \times 3} b^{4 \times 3} \\ &= -27 a^9 b^{12}\end{aligned}$$

$$\begin{aligned}\text{(c)} \quad \left(\frac{x^2}{y^4}\right)^3 &= \frac{(x^2)^3}{(y^4)^3} \\ &= \frac{x^{2 \times 3}}{y^{4 \times 3}} = \frac{x^6}{y^{12}}\end{aligned}$$

Summary of Positive Index Rules

If a and b are real numbers and m and n are integers, then

- $a^n \times a^m = a^{n+m}$
- $a^n \div a^m = a^{n-m}$
- $(ab)^n = a^n b^n$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$ (provided $b \neq 0$)
- $(a^n)^m = a^{n \times m}$